

# Replacing Phase 1 with Auction-Dynamics MBO

Approach B, measured against the validated deliverables — and a sixth occasion on which an instrument that could not see an effect reported its absence

August 2026

## Abstract

Phase 1 ( $\Gamma$ -convergence relaxation by projected gradient descent) costs 22 h–79 h for a completed  $N=300$  solution, and its winner-take-all readout is not a valid partition at that scale, so it is repaired afterwards. This report measures **approach B** — volume-constrained multiphase threshold dynamics [18, 9], with the volume constraint imposed by balanced assignment in the manner of auction dynamics [14] — as an outright *replacement* for Phase 1. Four results. (1) At the pinned Phase 2 campaign, B reaches perimeter **184.4118** and **184.1615** at  $N=100$  (two seeds) against the control’s 185.2546, and **319.9428** at  $N=300$  against 323.3192 — from 214–248 s of Phase 1 against 48,132 s and 79,069 s, a 211–319 $\times$  reduction in the replaced stage. (2) B produces **0 fragmented cells** at both  $N$  on its raw labels, where raw PGD at  $N=300$  produces 2 and requires the balanced readout; connectivity is the only one of the three validity gates that has content for B, and it is the one B could have failed. (3) The win is attributable to the MBO dynamics and not to the balanced geodesic initialisation B starts from — which is approach C’s first step — by +2.761% against a pre-registered 0.3% threshold. (4) Of thirteen predictions committed before any run, nine held, three missed and one was initially mis-scored. (5) At  $N=400$ , where no PGD anchor exists *or is possible*, B yields valid, connected, equal-area partitions at  $N=400$ , 500 and 750 from 920–3,614 s, agreeing with  $\sqrt{N}$  scaling on a fixed mesh to better than 0.06% across three points; at  $N=750$  the coarsest ladder rung falls *below* the freeze threshold and has to be dropped, which also removes the level-0 fragmentation seen at  $N=300$  and  $N=500$  — while on the *same* mesh the incumbent needed 206,344 s at  $N=300$  and still failed two of the three gates; the absent baseline is the finding, not a gap. The methodological content is again larger than the technical: a **three-part over-merge instrument was built and detected nothing**, and a per-level claim entered the record having never been measured — the sixth instance of this project’s signature artefact.

## Provenance

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status           | <b>measured</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Date             | 2026-08-19 (scored results); 2026-08-20 ( $N=400$ , exploratory); 2026-08-22 ( $N=400$ re-verification and export)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Surface          | torus $T(R=1.0, r=0.6)$ throughout                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Branch           | feat/phase1-mbo-auction-dynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Code             | src/partition/mbo_auction.py (the arm); scripts/run_mbo_arm.py (driver); scripts/score_mbo_arm.py (mechanical scorer); testing/test_mbo_auction.py (G1–G8, NC1–NC5b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Scored runs      | results/arm_mbo_20260819_173411_npart100_V114144_seed84172851, results/arm_mbo_20260819_185131_npart100_V114144_seed61803399, results/arm_mbo_20260819_192659_npart300_V47488_seed61803399, results/arm_init_20260819_181203_npart100_V114144_seed84172851                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Exploratory runs | results/arm_mbo_20260820_133709_npart400_V114144_seed84172851 results/arm_mbo_20260822_120646_npart500_V114144_seed84172851 and results/arm_mbo_20260822_182903_npart750_V158260_seed84172851 — $N=400$ , 500 and 750, <b>no anchor</b> , and none possible (§10). Scored against nothing; reported for gates, cost and absolute perimeter. The $N=500$ and $N=750$ Phase 2 campaigns were run with <code>--profile</code> . $N=750$ uses a <i>different</i> ladder ( $162 \times 154 \rightarrow 410 \times 386$ , $V=158,260$ ) and is therefore outside the fixed-mesh $\sqrt{N}$ comparison. $N=500$ predates the arm experiment.yaml snapshot, so it carries none; its config is parameters/torus_500part_mbo.yaml as of commit e8b8b60, which added an mbo: block after the run at the values the run used. |
| Anchors          | $N=100$ : run_20260709_081548, Phase 2 best <b>185.2546144718457</b> (iterate 20 of 20), raw PGD, Phase 1 48132.12s. $N=300$ : run_20260806_123326, Phase 2 best <b>323.319247087254</b> (iterate 18 of 19), PGD <i>plus</i> balanced readout, Phase 1 79068.56s. Both from timing_profile.yaml total_wall_s, never run_time_seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Pre-registration | commit 9ffdefb, 2026-08-19 16:53, <i>before</i> the first scored run at 17:34:11. Results commits b414b67, c8aea83, 5bdc947, 54b937c; corrections d4dc66d, d4b75d7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Figures          | make_figures.py from the committed data.yaml, itself re-derivable by extract_data.py.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Versions         | Python 3.12.13, numpy 2.4.4, scipy 1.17.1, h5py 3.16.0; macOS 15.3.1 arm64 (Mac mini).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Anchor check     | the harness reproduces the $N=100$ control’s own perimeter <i>bitwise</i> (report 07, 0d); no Phase 2 code has changed since.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## 1 What B is, and what is not claimed for it

Threshold dynamics originates with Merriman, Bence and Osher [18]: alternate short-time diffusion of the phase indicators with a pointwise reassignment, and interfaces move by approximate mean curvature. Esedoğlu and Otto [9] placed the multiphase case on a variational footing with a Lyapunov functional that  $\Gamma$ -converges to a weighted perimeter — the same limit this project’s own energy targets [3, 21, 20]. Jacobs, Merkurjev and Esedoğlu [14] impose *exact volume constraints* by replacing the pointwise threshold with a balanced assignment solved by Bertsekas’ auction algorithm [2]. **None of this is invented here.**

Per mesh level, the arm iterates on a *hard* labelling  $\omega : V \rightarrow \{0, \dots, N-1\}$ :

1. **diffuse**: solve  $(M + \tau K) y_k = M \chi_k$  for each cell, with one prefactorised LU per level and  $N$  back-substitutions;

2. **balanced threshold:**  $\omega(i) = \arg \max_k [y_{ik} + \psi_k]$ , with  $\psi$  from the shared semi-discrete optimal-transport dual [1, 17, 15] that approach A already ships.

There is never a continuous field whose winner-take-all readout can diverge from what was constrained. B therefore does not *close* the continuous–discrete gap that motivated approaches A–E; the category is absent.

**The regime, and the novelty claim we do *not* make. Corrected 2026-09-17.** An earlier version of this paragraph said that every published implementation of this machinery we were aware of convolves on a *uniform Cartesian grid via FFT*, on a flat domain, at small  $N$ , and for Pólya’s *outer* problem. That was written from two papers and was wrong on three of its four clauses — two falsified by one paper, the third by another. Reading the lineage itself (docs/math/10-mbo-auction-dynamics/ §2, all copies held at docs/papers/): Jacobs, Merkurjev and Esedoğlu [14, §4.2] already solve the *inner* problem — equal-area minimal-perimeter tessellation — of the *flat 2-torus*, a closed genus-1 surface, at  $N = 64$  (recovering the hexagonal lattice) and run the area-preserving flow of a 160-cell Voronoi diagram; Merriman and Ruuth [19, §8.3] evolve five regions of multiphase MBO on an embedded, *curved* torus and 175 regions on the sphere by the closest-point method; and Elsey, Esedoğlu and Smereka [7, 8] run diffusion-generated grain growth at  $10^5$  grains. Only “uniform grid via FFT” survives, and only for the Euclidean implementations — the graph line [10], whose implicit-Euler step is the one this arm transplants to surface finite elements, never used an FFT, and [14, §5] already run the *constrained* scheme on an unstructured similarity graph. Hu, Liu and Wang [13] remain the closest in *purpose* (auction dynamics as the inner solver of the outer problem), at  $n \leq 9$  on  $256^2/128^3$  grids, and Bogosel and Oudet [4] use capacity-constrained Voronoi diagrams with  $> 100$  planar cells as an initialisation.

What the held literature does not contain, stated as “we are not aware of” and never as “first”: the scheme on a curved embedded surface discretised by an unstructured triangle mesh with a surface-FEM diffusion step; equal-area constraints at  $N = 400$ –1000; an inexact transportation dual in place of the exact auction; and the multilevel mesh ladder. The defensible framing is *combination, surface and scale*, not *method*. The corpus is now the twenty-odd papers listed in docs/papers/README.md, read at the sections cited, plus this project’s bibliography; it is still not a systematic review, and surface-remeshing and computer-graphics venues may cover adjacent ground under different vocabulary.

The difference is load-bearing rather than cosmetic. The FEM operator  $(M + \tau K)^{-1}M$  is the *resolvent* of the Laplace–Beltrami operator, not a Gaussian convolution: document 10 derives that one step is mean-curvature flow over time  $\tau/2$  rather than  $\tau$ , that  $E_\tau$  tends to  $\frac{1}{2} \sum_k \text{Per}(\Omega_k)$  rather than  $\pi^{-1/2}$  times it, and checks the former on the production mesh; mesh pinning [23] is ours to manage, and no grid-derived  $\tau$  heuristic is imported.

## 2 Result 1 — perimeter

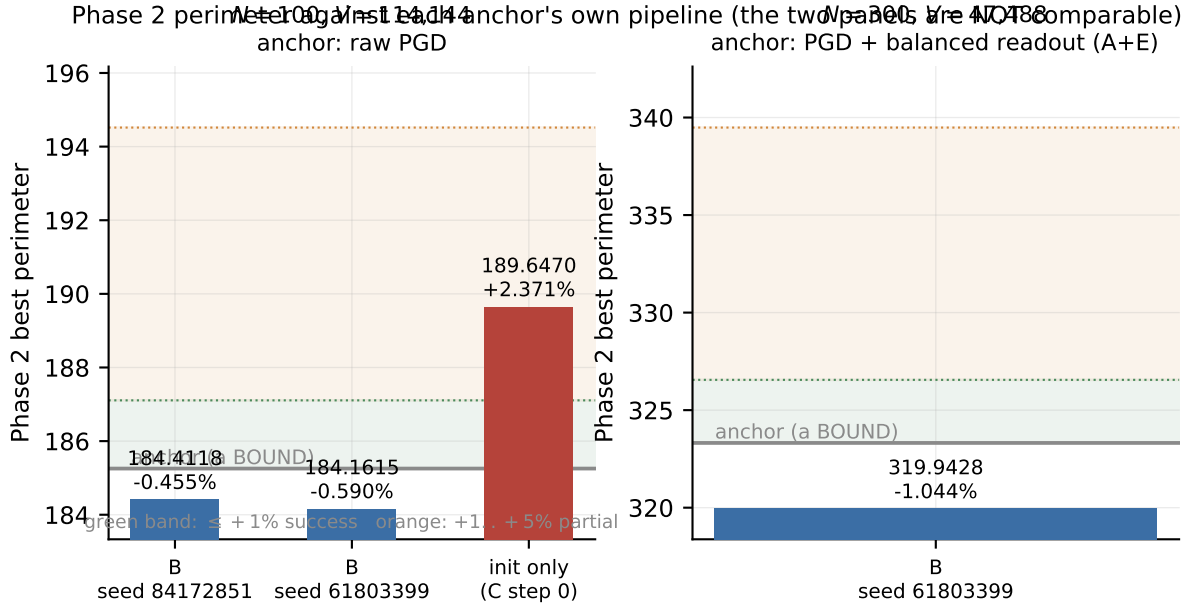


Figure 1: Phase 2 best perimeter against each anchor’s own pipeline. **The two panels are not comparable and must never be pooled:** the  $N=100$  anchor is raw PGD, the  $N=300$  anchor is PGD *plus* the balanced readout, because raw PGD is not a valid partition at  $N=300$ .

| run                  | Phase 1 | speedup | area worst | frag.    | Phase 2 best              | vs anchor       |
|----------------------|---------|---------|------------|----------|---------------------------|-----------------|
| $N=100$ s84172851    | 228 s   | 210.9×  | 0.1504%    | <b>0</b> | 184.4117703525215 (18/20) | − <b>0.455%</b> |
| $N=100$ s61803399    | 214 s   | 225.2×  | 0.1493%    | <b>0</b> | 184.1614647856558 (19/20) | − <b>0.590%</b> |
| $N=300$ s61803399    | 248 s   | 318.8×  | 1.4172%    | <b>0</b> | 319.9427628182375 (16/19) | − <b>1.044%</b> |
| init only (C step 0) | 67 s    | —       | 0.1355%    | 0        | 189.6469917457051 (20/20) | +2.37%          |

Seed spread at  $N=100$  is **0.136%**, which is  $3.3\times$  smaller than the margin by which the *worse* of the two seeds beats the control. This is what makes the result scoreable at all: the pre-registration committed that a spread above 1% would mean the primary falsifier could not be evaluated at  $\pm 1\%$  and that a single-seed number would be conflating method difference with seed noise.

**Six constraints on how these numbers may be stated.**

1. **Never pool the two anchors.** They are different baselines. “B beats the best available PGD pipeline at each  $N$ , at equal Phase 2 budget” is valid; “B beats PGD by 0.455–1.044%” as a trend in  $N$  is not.
2. **The speedup is for the replaced stage only.** End-to-end, including the shared 30–40 min Phase 2, it is  $\approx 23\times$  ( $N=100$ ) and  $\approx 31\times$  ( $N=300$ ). Against the structure-trigger PGD variant (34,967 s), the incumbent-best figure is  $\approx 153\times$ .
3. **Neither trajectory is a converged optimum.** Both are at the documented migration-cycling plateau. An earlier draft said “B converged where the control had not”; the sixth review resumed the control’s own Phase 2 for **20 further iterations and found no improvement** (best over all 40 remains 185.2546 at iterate 20; the extension drifts up to  $\approx 185.30$ ). B’s plateau is simply 0.455% lower. *That extension is the review’s measurement*

and has not been independently reproduced by us.

4. **PGD’s own seed variance is unmeasured.** Every  $N=100$  PGD run across both working directories is seed 84172851, and every  $N=300$   $\lambda=11.5$  run is seed 61803399. Each anchor is a single trajectory. “B beats PGD by more than PGD’s own seed lottery” is *not* established.
5. **The  $N=300$  margin cannot be decomposed, even in principle.** The readout costs +1.882% in label-boundary length (189.7310 raw  $\rightarrow$  190.7114 shifted  $\rightarrow$  193.3024 repaired, the last being what the anchor’s Phase 2 consumed). B’s raw boundary is 187.2574: shorter than the repaired labels by 3.127% *and* shorter than raw PGD’s own invalid labels by 1.304% — the strong form, since imbalance can only make a boundary artificially short. So the margin is not merely “not paying the repair bill”. But how much is un-erased repair residue cannot be separated, because **no valid repair-free PGD partition exists at  $N=300$ .**
6.  $N=300$  rests on one Phase-2-scored seed.

### 3 Result 2 — validity, and which gate actually has content

“All three gates pass” is true and misleading

An earlier write-up of this result — ours — said B’s raw labels pass all three validity gates. Phase 0’s own findings say why that is the wrong emphasis:

- **Dormant is vacuous** for any balanced-assignment arm. One-hot labels give peak density 1.0 by construction; the gate cannot fail. The harness already emitted `vacuous_for_arms: True`.
- **Area is near-expected.** Equal area is precisely what B’s assignment step *optimises*. Passing means the solver converged; Phase 0 defined it as the *solver-failure detector*, not a validity test.
- **Connectivity is the only gate genuinely at risk.** Nothing in B enforces it, and the source proposal states it is “not per-step guaranteed”.

The defensible claim is narrower and still strong: **B produces 0 fragmented cells at  $N=300$ , where raw PGD produces 2** (with 10 imbalanced cells, worst 36.15%) — which is why PGD’s baseline required the balanced readout and B required nothing.

Re-verified through `testing/check_fragmentation.py`, which shares no code with the driver’s reporting: all four arm solutions pass, with **0 cells having more than one raw component including sub-threshold speckle**.

**Robustness.** Three further  $N=300$  MBO ladders at seeds 27182818, 13131313 and 84172851 give **0 fragmented** in every case, with worst area 1.2531/1.4467/1.3887% and label boundaries 186.08/186.55/186.90 — all *shorter* than the scored seed’s 187.2574. Ladder only, no Phase 2; *review-produced, not independently reproduced by us*.

### 4 Result 3 — attribution: is this B, or is it C’s first step?

B is initialised from one balanced C-scores assignment — farthest-point seeds, multi-source Dijkstra, scores  $-d^2$ , balanced assignment — which is literally approach C’s step 0 [16, 1, 6]. If MBO barely improved on it, the number in §2 would be a result about C reported under B’s name, and *nothing* in the gate suite would have detected that.

|                      | Phase 1  | label boundary | Phase 2 best                     | censored   |
|----------------------|----------|----------------|----------------------------------|------------|
| init only (C step 0) | 67 s     | 116.1989       | 189.6469917457051 (20/20)        | <b>yes</b> |
| <b>B (MBO)</b>       | 228 s    | 107.6014       | <b>184.4117703525215</b> (18/20) | no         |
| control (PGD)        | 48,132 s | 107.3410       | 185.2546144718457 (20/20)        | plateaued  |

$$\text{P13} = \frac{189.646992 - 184.411770}{189.646992} = +\mathbf{2.761\%} \quad \text{against a pre-registered threshold of 0.3\%}.$$

Two disclosures. The measured 2.761% is an **upper bound** on MBO’s contribution, because the init arm is censored: its converged value is  $\leq 189.6470$ , and a lower true value shrinks the gap. For attribution to fail the init would have to reach  $\approx 185.0$  — another  $\approx 2.5\%$  on a trajectory gaining  $\approx 0.038$  per iterate and decelerating. And the init-only control was run on the *finest* mesh, i.e. the *strongest* member of the init family, which biases the test *against* B.

**An unflattering free datum on C.** C’s step 0 plus Phase 2 lands at +2.37% above the PGD control — a PARTIAL verdict under this project’s own bands — for 67 s of Phase 1. Real C iterates Lloyd and would improve on this, so it is a *lower bound* on C; but C’s starting point is behind both B and PGD, which is directly relevant to whether C is worth building. Note also that C minimises a *quantisation* energy, not perimeter; the two optima coincide only asymptotically and only in the plane [12, 11], and this torus has Gaussian curvature of both signs.

## 5 The mesh ladder, and $\tau$

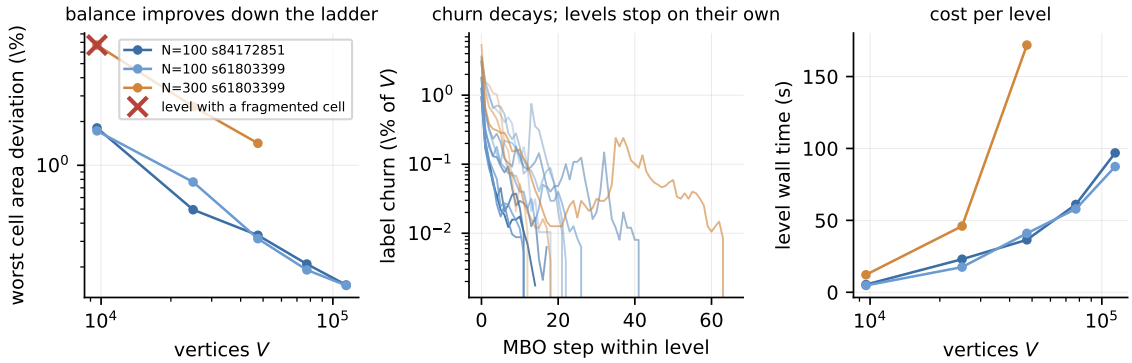


Figure 2: Per-level behaviour. Left: worst cell area deviation improves monotonically down the ladder; the cross marks the one level carrying a fragmented cell. Middle: label churn decays within every level, so levels terminate on their own rather than at the iteration cap. Right: cost per level.

$\tau = \min((ch_{\text{mean}})^2, (\rho R_{\text{cell}})^2)$  with  $c = 4$  and  $\rho = 1.0$ , pinned before any run.  $c = 4$  rather than  $c = 2$ : the latter is the freeze regime [23]. The cap is the over-merge side of a two-sided window, and it binds on exactly one level in use,  $N=300$  level 0.

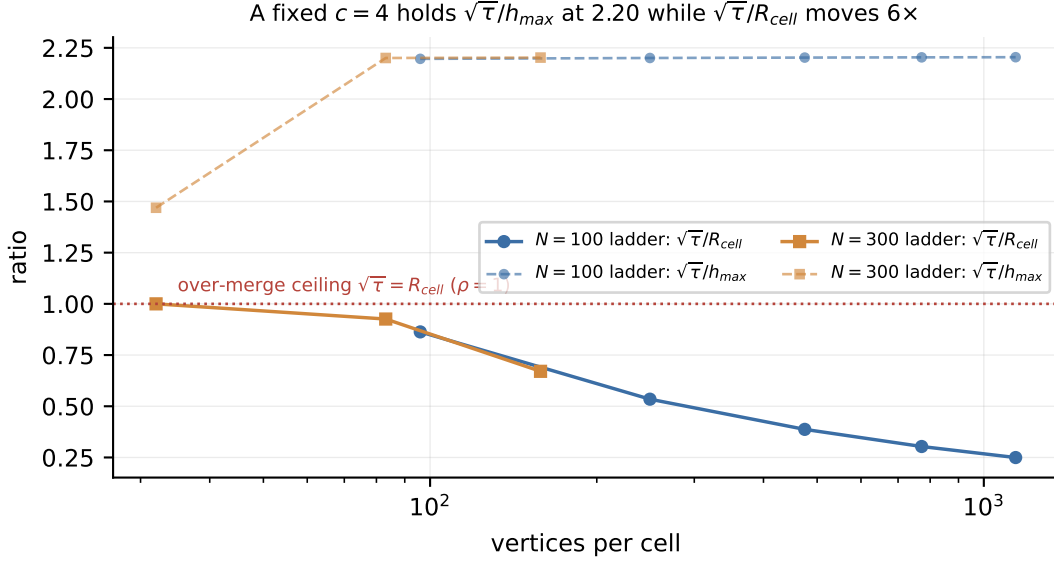


Figure 3: A fixed  $c$  does *not* hold the window.  $\sqrt{\tau}/h_{\max}$  stays at 2.20 across every level of both ladders while  $\sqrt{\tau}/R_{\text{cell}}$  moves  $6\times$ . Reporting on  $h_{\text{mean}}$  alone flattens the coarse band: this mesh is  $1.81\times$  anisotropic.

## 6 The headline null: an instrument that detected nothing

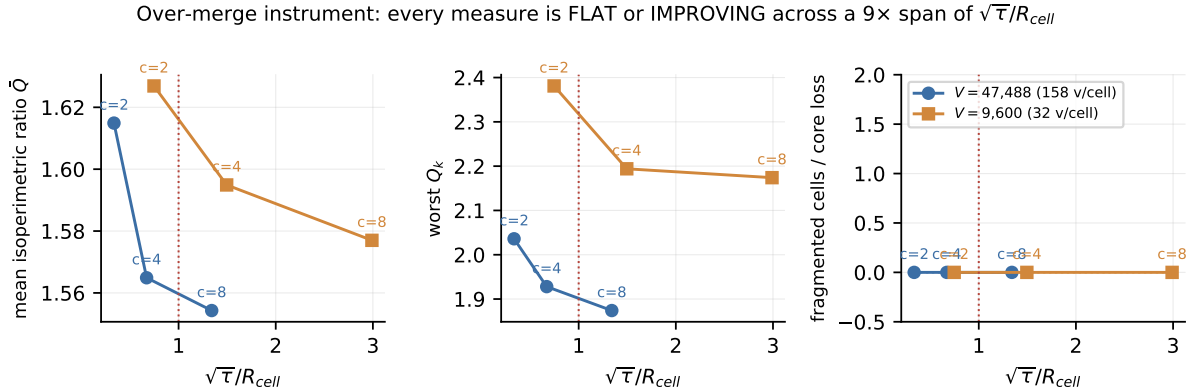


Figure 4: Fragmentation, per-cell isoperimetric ratio  $Q_k = P_k^2/(4\pi A_k)$ , and core loss, swept over  $c \in \{2, 4, 8\}$  at two resolutions. Every measure is flat or *improving*; none registers an over-merge penalty anywhere, including at  $\sqrt{\tau}/R_{\text{cell}} = 2.99$ , three times the pinned cap.

Assignment quality *cannot* see over-merging — the balanced assignment hits its area target by construction, so a smeared, non-local cell scores as well as a compact one. (An earlier draft of the plan cited a healthy assignment-quality fixture as evidence against over-merging; that was a measurement of a different quantity than the claim it supported, and it was withdrawn.) So three geometry instruments were built. Across a  $9\times$  span of  $\sqrt{\tau}/R_{\text{cell}}$  and two levels differing  $5\times$  in resolution, **not one of them registered anything**: core loss and fragmentation are 0 in all six configurations, and compactness moves monotonically the *other* way.

The instruments are demonstrably *responsive* —  $\bar{Q}$  moves 3.8%,  $Q_{\text{worst}}$  8%, boundary length 1.9% — they simply move toward improvement. What is established is therefore that **no damage is**

**detectable**, *not* that over-merging cannot occur.

#### Standing limit

The  $\rho$  question is **open**, and no future claim about  $c$  or  $\rho$  may lean on this report’s evidence in either direction. The cap was retained because it is pre-registered and because it biases *against* B — it lowers  $c$  to 2.68 at  $N=300$  level 0, and  $c=2$  measures worse than  $c=4$ . That is a discipline argument, not an empirical one. **Named follow-up:**  $c=8$  showed no penalty on any instrument *and* a 0.33% shorter starting boundary — a third of the entire +1% success bar — so the cap may be costing perimeter for no measured benefit. That is a separate arm, scored through the same harness.

## 7 Why there is no exactly-monotone energy, and what is gated instead

Esedoglu–Otto monotonicity is a minorise–maximise argument requiring (i) the same symmetric positive semi-definite form in the energy and in the assignment objective, and (ii) an *exact* assignment. **Both fail here, for independent reasons.** The theorem’s object is  $MA_\tau = M(M + \tau K)^{-1}M$ , symmetric PSD; our step maximises  $\langle \chi', Dy \rangle$  with  $D = \text{diag}(v)$  *lumped*, and  $DA_\tau$  is not symmetric. And Jacobs, Merkurjev and Esedoglu obtain a theorem because Bertsekas’ auction is exact, whereas `solve_dual_offsets` is subgradient ascent returning its best iterate. **This is the price of a substitution made deliberately in Phase 0** — where Sinkhorn [5] and auction were both measured and both lost to the incumbent on assignment quality — paid here rather than there.

What survives needs neither concavity nor exactness. Since  $\omega'(i) = \arg \max_k [y_{ik} + \psi_k]$ , multiplying the pointwise optimality by  $v_i > 0$  and summing gives

$$\langle \chi', Dy \rangle - \langle \chi, Dy \rangle \geq \sum_k \psi_k (T_k - T'_k)$$

whose right-hand side is exactly the slack that inexactness costs; it vanishes at perfect balance. Measured slack runs  $1.7 \times 10^{-4}$  to  $5.5 \times 10^{-3}$  — five to seven orders of magnitude above the  $10^{-9}$  tolerance an earlier draft proposed for energy monotonicity, so that tolerance was never defensible. **Real  $E_\tau$  increases do occur** (4 of 96 active steps at  $N=300$ ), and gate G4 therefore gates the inequality, whose measured violation is  $\leq 6.13 \times 10^{-8}$ , while  $E_\tau$  is reported as a descriptive trend only.

## 8 Predictions, and what they caught

|            | PREDICTED (committed 9ffdefb)        | MEASURED                        |                            |
|------------|--------------------------------------|---------------------------------|----------------------------|
| <b>P1</b>  | 188.0 (+1.5%), PARTIAL               | 184.4118 / 184.1615, SUCCESS    | <b>MISS</b>                |
| <b>P2</b>  | spread < 0.5%                        | 0.136%                          | <b>HELD</b>                |
| <b>P3</b>  | <= 0.28%, point 0.15%                | 0.1504 / 0.1493%                | <b>HELD</b>                |
| <b>P4</b>  | <= 3, point 0                        | 0                               | <b>HELD</b>                |
| <b>P5</b>  | >= 30x, point 53x                    | 210.9 / 225.2x                  | <b>HELD (point missed)</b> |
| <b>P6</b>  | 331.4, 80% CI [323.3, 341.0]         | 319.9428                        | <b>MISS</b>                |
| <b>P7</b>  | <= 2.02%, <= 6 frag, >= 88x          | 1.4172%, 0, 318.8x              | <b>HELD</b>                |
| <b>P8</b>  | <= 2 of 5 flagged                    | 0, clauses agree                | <b>HELD</b>                |
| <b>P8'</b> | dE flags L2,L3,L4 (falsifying P8)    | dE flat 14-22, flags none       | <b>MISS</b>                |
| <b>P9</b>  | frag not death; worst at L0          | 1 frag at L0 (295), 0 elsewhere | <b>HIT</b>                 |
| <b>P10</b> | viol <= 1e-12; excursions <= ceiling | -6.13e-08; 4/96 within ceiling  | <b>HELD</b>                |
| <b>P12</b> | no abort                             | 20/20, 19/19                    | <b>HELD</b>                |
| <b>P13</b> | >= 0.3%                              | +2.761%                         | <b>HELD</b>                |

Figure 5: Thirteen predictions committed in 9ffdefb at 16:53, before the first scored run at 17:34:11.

**P1 and P6 missed, and P1’s *reasoning* is refuted.** The rationale was that B sees perimeter only through a coarse- $\tau$  surrogate while PGD saw the actual energy for 13.4 h. B also *enters* Phase 2 from a 0.243% *longer* boundary than the control and still finishes ahead — a genuine crossover, so the advantage is not initial geometry. That it is combinatorial is a hypothesis consistent with these data, not established by them. P6 missed worse: its 80% interval’s lower bound was the anchor itself, so essentially no probability was assigned to B beating PGD + A + E at  $N=300$ .

**P8’ missed and invalidated its own model.** It predicted, from a log-linear interpolation of  $\Delta E/\text{tail}$  in the freeze ratio  $c/c_{\text{lo}}$ , that levels 2–4 would be flagged pinned. Measured,  $\Delta E/\text{tail}$  is *flat* across the ladder — 15.76, 21.62, 15.11, 14.22, 16.49 against a threshold of 79.25 — and level 4, sitting exactly at  $c/c_{\text{lo}} = 0.999$ , scores 16.49 where the model demanded  $\approx 260$ . The freeze ratio does not predict pinning here.

## 9 The sixth artefact — and it is ours

Report 07 catalogued five occasions on which a measurement artefact was read as a result about the method. This is the sixth, and it was produced in this work.

The plan asserted “**0 fragmented at every level**” and, on that basis, scored P9 — which predicted over-merge would appear as fragmentation rather than cell death, worst at level 0 — as UNTESTABLE, “the predicted phenomenon never occurred”. **Until the sixth review, LevelReport carried no connectivity field at all.** Per-level fragmentation had never been computed. The claim was an inference presented as a result, and it was false.

With the instrument added and the  $N=300$  ladder re-run to a bit-exact reproduction (worst areas 6.7063/2.5394/1.4172%, steps 16/21/66, boundary 187.2574):

**level 0: 1 fragmented cell (cell 295). Levels 1 and 2: 0. No cell death.**

The phenomenon occurred, at exactly the level P9 named, in exactly the form P9 predicted. **P9 is a hit, not untestable** — “untestable” had absorbed a correct prediction on the strength of a measurement never taken. The stray is healed by level 1’s MBO run, which is direct evidence for the mechanism the taxonomy predicted: stray islands are perimeter-inefficient and shrink under curvature flow, the opposite dynamic to the discrete-area trim that manufactured them.

The fix is instrumentation — `run.gates` per level, seconds of cost, now committed — not a resolution to be careful.

## 10 Result 5 — $N=400, 500, 750$ : existence where the anchor cannot exist

The three results above are all *margins*: B against a control, at a pinned Phase 2 budget. That design has a ceiling built into it. It can only be run where a control exists, and the control is the method whose cost is the reason B was built. Pushed far enough in  $N$ , the comparison stops being expensive and starts being impossible.

$N=400, 500$  and  $750$  are past that point, and this section reports them as what they are: **existence-and-validity** results with no anchor, no margin, and no comparison of perimeters. All three ran in the driver’s exploratory `--config` mode, where the mesh-match assertion is skipped because there is nothing to match.  $N=500$  deliberately reuses  $N=400$ ’s ladder unchanged, which converts the  $\sqrt{N}$  argument below from a single check into a three-point one;  $N=750$  could not, for a reason that is itself a result (§10).

### The results

$N=400$  and  $N=500$  run to  $V = 114,144$ ;  $N=750$  runs to  $V = 158,260$ . All three validity gates pass on the *raw* labels at every  $N$ , re-verified with `testing/check.fragmentation.py` independently of `arm.report.yaml`:

|                      | $N=400$                        | $N=500$                        | $N=750$                        |
|----------------------|--------------------------------|--------------------------------|--------------------------------|
| Phase 1 wall         | 920 s                          | 1,322 s                        | 3,614 s                        |
| dormant              | 0 / 0                          | 0 / 0                          | 0 / 0                          |
| area imbalance (bar) | 0, <b>0.7773%</b> (1.1214)     | 0, <b>1.0993%</b> (1.4017)     | 0, <b>1.2569%</b> (1.5165)     |
| connectivity         | <b>0</b>                       | <b>0</b>                       | <b>0</b> at <i>every</i> level |
| Phase 2 perimeter    | <b>368.6603</b><br>(it. 19/20) | <b>412.1138</b><br>(it. 10/20) | <b>504.9214</b><br>(it. 14/20) |
| variable points      | 29,287                         | 32,704                         | 47,088                         |
| max area violation   | 8.18e−7                        | 1.10e−6                        | 4.68e−7                        |

All three Phase 2 constraint residuals are converged to tolerance, so the partitions are genuinely equal-area rather than approximately so, and the  $N=500$  and  $N=750$  bests are *mid-trajectory* — not censored; both descend and then oscillate in the migration-cycling plateau. All three deliverables are exported with `finalised=True`.

### The $\sqrt{N}$ cross-check, three points

Not a gate, and the sharpest instrument available in the absence of an anchor. On a *fixed* mesh total perimeter should scale as  $\sqrt{N}$ ; all three runs below sit on the same  $V = 114,144$

mesh, so the comparison is free of any refinement effect.  $N=500$  was given  $N=400$ 's ladder unchanged precisely to make this test available, and its predicted value was recorded in `parameters/torus.500part_mbo.yaml` *before* the run.

|                           | predicted | measured        | deviation      |
|---------------------------|-----------|-----------------|----------------|
| $N=100 \rightarrow N=400$ | 368.8236  | 368.6603        | -0.044%        |
| $N=400 \rightarrow N=500$ | 412.1748  | <b>412.1138</b> | <b>-0.015%</b> |
| $N=100 \rightarrow N=500$ | 412.3573  | 412.1138        | -0.059%        |

Three independent runs at three values of  $N$  agree with the scaling to better than 0.06%. All three deviations are negative, which is a consistent slight shortfall rather than scatter; with three points we do not attempt to say why.

$N=750$  is *not* in that table, because it runs on a different mesh and the test is only sharp at fixed  $V$ . Recorded for completeness and scored against nothing: the fixed-mesh value implied by  $N=500$  is 504.7342 and the measurement is 504.9214, +0.037%. Two points worth stating plainly. The magnitude is small, but the *sign* is opposite to all three fixed-mesh checks, and the pre-registered expectation in `parameters/torus.750part_mbo.yaml` was that a finer mesh should land slightly *below* — so the stated direction was wrong. Nothing is concluded from a single unscored point; it is noted so the error is on the record.

### The ladder had to change at $N=750$ , and the diagnostics said so first

Every earlier config began at  $100 \times 96$ . At  $N=750$  that rung is unusable, and this is visible *before* running: it gives 12.8 verts/cell and  $\sqrt{\tau}/h_{\max} = 0.929$  against a local non-freeze threshold of 0.964 — a ratio of **0.964**, **below 1**, so the coarse-triangle band cannot move at all. Every measured run had been above 1 at every level ( $N=400$  min 1.049,  $N=500$  min 1.067).

The mechanism is a squeeze that tightens with  $N$ : more cells means smaller cells, the over-merge cap holds  $\sqrt{\tau} \leq R_{\text{cell}}$ , so the diffusion length shrinks — while the coarse mesh's triangles do not. Past some  $N$  the diffusion no longer reaches across one triangle and the level freezes. **The remedy is the ladder, not  $\tau$** : raising  $\tau_c$  walks toward the over-merge wall on the other side of the same two-sided window. The config therefore keeps levels 1–5 verbatim and drops the dead rung, running  $162 \times 154 \rightarrow 410 \times 386$ . Every level then sits at ratio  $\geq 1.131$ , and the finest level's margin (1.131) is better than  $N=500$ 's (1.109) and  $N=400$ 's (1.049), both of which ran clean.

It worked, and it bought something beyond avoiding the freeze:  $N=750$  is the first high- $N$  run with **zero fragmented cells at every level**, where  $N=300$  and  $N=500$  each produced one at level 0 that later levels had to heal. Starting at 33 verts/cell instead of 12.8 or 19.2 removes the defect rather than repairing it.

### Two mechanisms recur at a third $N$

Neither is new; both were predicted or observed earlier, and  $N=500$  is an independent occasion for them to fail.

**Level-0 fragmentation, healed by level 1.**  $N=500$  level 0 produces one fragmented cell (323) and level 1 has none — exactly the pattern §9 recorded at  $N=300$  (cell 295), and the mechanism the taxonomy predicts: stray islands are perimeter-inefficient and shrink under curvature flow.  $N=400$  shows none at any level. This is visible only because the per-level instrument now exists; it is the same claim that was once asserted with nothing measuring it.

**The over-merge null holds.** At  $N=500$  the cap binds on two levels and  $\sqrt{\tau}/R_{\text{cell}}$  is at its ceiling there, yet the instrument reports core loss = 0 at every level,  $q_{\text{worst}}$  falling  $2.95 \rightarrow 1.81$

down the ladder, and  $q_{\text{mean}}$  in 1.54–1.61 against  $N=400$ ’s 1.56–1.57. A third  $N$ , a tighter cap, and the same null of §6. Still undetected rather than excluded.

### The mesh-matched companions, and a level 0 below the freeze threshold

The runs above each use the finest ladder their  $N$  can support, so  $N=750$  and  $N=1000$  end at  $V=158,260$  and  $209,568$  while every earlier deliverable ends at  $114,144$ . The exported partition carries the Phase 1 mesh, so a downstream study measuring per-cell quantities cannot compare across  $N$  when the mesh changes underneath it. Both were therefore re-run on the *common* mesh — and not merely to the same endpoint. Matching only the endpoint would be a different algorithm: in a multi-level scheme the coarse levels do the structural work and the fine levels refine it. The companions use the **identical ladder, seed and  $\tau$  constants as the  $N=400/500$  configs**, changing only `n_partitions`.

That forces  $100 \times 96$  as the base rung, where the two large  $N$  sit *below* the freeze threshold — a regime no previous run had entered:

|          | level-0 ratio | v/cell | level-0 floor | final worst    | bar     |
|----------|---------------|--------|---------------|----------------|---------|
| $N=750$  | 0.964         | 12.8   | 12.4997%      | <b>1.6928%</b> | 2.1026% |
| $N=1000$ | 0.897         | 9.6    | 16.6663%      | <b>2.2560%</b> | 2.8035% |

**Both pass all three gates: 0 dead, 0 weak, 0 imbalanced, 0 fragmented.** So a sub-threshold level 0 is *survivable* under B. Three observations, the first two of which were not predicted:

(1) **It is a partial, not total, freeze — and it costs real work.** Scoring each level by the fraction of the improvement available to it that it actually captured (its floor being the target it cannot pass), level 0 captures 18% at  $N=750$  and 26% at  $N=1000$ , where every healthy level in both runs captures 55–89%. About a third of normal. This matches the mechanism: the diffusion reaches past the mean edge but not the largest, so fine-triangle regions evolve while the anisotropic coarse band does not.

(2) **The freeze ratio does not predict how much a sub-threshold level accomplishes.**  $N=750$  has the *better* ratio (0.964 vs 0.897) and more verts/cell, and was predicted to fare at least as well; it captured *less* (18% vs 26%). With two points no mechanism is offered, but the ratio alone is evidently not the controlling variable below 1.0.

(3) **The damage is manufactured at level 0 and fully healed.** Level 0 produced 2 fragmented cells at  $N=750$  and 3 at  $N=1000$  — more than any run above threshold — and the chains clear completely:  $2 \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow 0$  and  $3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 0$ . This is the distinction from PGD, where `docs/experiments/06-subfloor-ladder/` measured a starved level 0 leaving a *permanent* runt (46.80%  $\rightarrow$  15.92% across  $37\times$  more vertices). Under B the same insult is transient.

### Two properties of the common mesh worth recording

**Worst-cell error saturates at  $\approx 1.61\times$  the granularity floor.** On the fixed  $V=114,144$  mesh:

|          | worst   | floor   | ratio        |
|----------|---------|---------|--------------|
| $N=400$  | 0.7773% | 0.5607% | 1.386        |
| $N=500$  | 1.0993% | 0.7009% | 1.568        |
| $N=750$  | 1.6928% | 1.0513% | <b>1.610</b> |
| $N=1000$ | 2.2560% | 1.4017% | <b>1.609</b> |

It rises and then plateaus. If that holds, quality on a fixed mesh is predictable from the mesh alone — worst cell  $\approx 1.61\times$  granularity, which scales as  $N/V$  — so a mesh can be sized for a target area error without running anything. Four points and no mechanism: a pattern, not a law.

**A coarser mesh under-measures perimeter.**  $N=1000$  reads 583.1417 at  $V=114,144$  against 583.2558 at 209,568,  $-0.020\%$ . The direction is not noise: a boundary represented by fewer, longer straight segments is a shorter approximation to the same curve, so the coarse value is an underestimate and the finer one is closer to the truth. Small, but it biases any perimeter-derived quantity systematically with  $N$  when the mesh is held fixed — which is the price of holding it fixed.

**Equal-area accuracy of the exported geometry is unaffected by any of this.** The percentages above are Phase 1’s *discrete* winner-take-all imbalance and do not survive into the export; Phase 2 equalises the actual geometric areas. Measured on the exported best iterates, the worst cell is off target by 0.0014% ( $N=400$ ), 0.0023% (500), 0.0272% (750) and 0.0127% (1000) — the two mesh-matched runs are the loosest of the set, as fewer verts/cell would predict, and all four are negligible.

### Why the missing anchor is the finding

The obvious objection is that a result without a baseline is weak. Here it is the reverse, and the reason is measurable rather than rhetorical. Compare at the **matched** finest mesh  $V = 114,144$  — the same mesh, so no refinement effect enters:

|                      | PGD, $N=300$<br>run_20260808_191030 | B, $N=400$<br>arm_mbo_20260820_133709 |
|----------------------|-------------------------------------|---------------------------------------|
| Phase 1 wall         | 206,344 s (57.3 h)                  | <b>920 s</b>                          |
| dead / weak          | 0 / 0                               | 0 / 0                                 |
| area imbalance       | <b>3 imbalanced, worst</b> 24.81%   | 0, worst <b>0.78%</b>                 |
| fragmented           | <b>2</b>                            | <b>0</b>                              |
| valid on raw labels? | <b>no</b> — requires the readout    | <b>yes</b>                            |

On the identical mesh, with *one hundred fewer cells*, the incumbent took **224** $\times$  the wall time and failed two of the three gates. Requiring a PGD anchor at  $N=400$  therefore asks the incumbent to do precisely the thing whose absence motivated B. The honest claim needs no margin:

Approach B produces a valid, equal-area, fully connected 400-cell partition on a 114,144-vertex torus from 15 minutes of Phase 1, in a regime where the incumbent has already ceased to produce valid partitions at  $N=300$  on the same mesh.

### What this section does *not* claim

The  $-0.455\%$  and  $-1.044\%$  margins of §2 are not extended to  $N=400$  or  $N=500$ , and no perimeter here is compared against PGD. The matched-mesh table above is a statement about *validity and cost*, both measured on both sides at the same  $V$ ; it is not a perimeter comparison, and the two must not be blurred. The  $\sqrt{N}$  check is B against B, so it tests internal consistency, not correctness against an independent method: three runs agreeing on a scaling law they all obey would not detect an error common to all three. And  $N=500$  reuses  $N=400$ ’s ladder and seed, so the two are not independent in the mesh or the initialisation, only in the run.

### The cost centre has moved — and it is not the solver

Phase 2 took 7,326 s, 8,829 s and **18,323 s** (5.09 h) at  $N=400/500/750$ , against Phase 1’s 920 s, 1,322 s and 3,614 s. The stage this report replaces is therefore 11%, 13% and 16.5% of end-to-end

cost — the last rising only because  $N=750$  was given an extra ladder level. The 211–319× of §2 was always a statement about the replaced stage; past  $N=400$  the remainder dominates.

The  $N=500$  and  $N=750$  campaigns were run with `--profile`, and the breakdown is not what the forward plan assumed. **The IPOPT solver accounts for 5.2% of the campaign at  $N=500$  and 3.6% at  $N=750$  — a *shrinking* share of a growing problem.** At both, IPOPT runs 201 solver iterations per topology iteration, hitting `max_opt_iter: 200` every time, and the internal callback split is essentially identical (Hessian 40.4% vs 40.7% of solver time). So 95–96% of Phase 2 is spent *outside* `optimize()`: contour rebuild, Steiner setup, migration detection and checkpoint export with its roundtrip verification. **We establish the solver’s share, not the composition of the remainder** — decomposing that is the measurement docs/plans/PUBLICATION\_READINESS\_PLAN.md Phase 4 should now make, in place of its original premise that the exact-Hessian path was the target.

**Phase 2 also scales superlinearly in variable points, close to quadratically.** Campaign wall against variable-point count:

|         | variable points | wall     | s/VP  | exponent    |
|---------|-----------------|----------|-------|-------------|
| $N=400$ | 29,287          | 7,326 s  | 0.250 |             |
| $N=500$ | 32,704          | 8,829 s  | 0.270 | 1.69        |
| $N=750$ | 47,088          | 18,323 s | 0.389 | <b>2.00</b> |

$N=1000$  has since been measured and extends the table: 62,472 variable points, 29,946 s, 0.479 s/VP, exponent **1.74**. **An earlier version of this paragraph called the trend “superlinear and steepening”; the fourth point refutes the second half** — the exponents run  $1.69 \rightarrow 2.00 \rightarrow 1.74$ , so 2.00 was an excursion, not a trend. The defensible statement is *superlinear, exponent  $\approx 1.7$ – $2.0$ , with no evident trend*, which is a better outlook than the wording it replaces because it means the cost is not accelerating away.

One cleaner comparison is now available.  $N=750$  and  $N=1000$  both have mesh-matched companions on  $V=114,144$  (§10), giving the first pair measured at *fixed* mesh: 39,704 VPs / 12,908 s and 45,859 VPs / 18,138 s, an exponent of **2.36** — steeper than any of the mixed-mesh figures, which suggests part of what was being attributed to variable-point count was a mesh effect. Two points, so this is a flag for a third measurement rather than a result. Either way Phase 2 is the binding constraint on  $N$ .

*Measurement note.* The  $N=750$  mesh-matched campaign was launched twice — the first attempt wrote iterate 1 and stopped — leaving 21 iterate files, two of them bit-identical copies of iteration 1. Its wall time must be taken from the *second* launch (12,908 s); measuring from the campaign directory spans the aborted attempt and reads 14,021 s, 9% high. It is the only campaign of the ten with more than one launch.

#### A reporting trap in `timing_profile.yaml`

The Phase 2 profile’s `summary.total_wall_s` is 459.27 s while the campaign took 8,829 s. The field accumulates only time inside `optimizer.optimize()`; everything else in the loop is outside it. A **19×** under-report if read as campaign wall — the same class of trap as `run_time.seconds` in the Phase 1 metadata, which under-reports by 3.6× and is already documented. Campaign wall must be taken from the iterate timestamps.

Two scaling consequences worth recording. The  $\tau$  over-merge cap binds on *two* levels at both  $N$  (against one at  $N=300$  and none at  $N=100$ ) as cells shrink relative to the coarse mesh. And the vertex-granularity floor has risen from 0.5607% to 0.7009%, pushing the worst cell from 1.39× to 1.57× that floor against a 2× bar — so the finest mesh will have to grow before the bar is reached, and that raises Phase 2, which is now the expensive half.

## 11 Threats to validity that no instrument here addresses

1. **PGD’s seed variance** (§2, constraint 4). Both anchors are single trajectories; the corpus was enumerated across *both* working directories, since `results/` is gitignored and each worktree has its own.
2. **Over-merging** (§6). Not excluded, merely undetected.
3. **Phase 2 budget as a confound**. B may start from geometry that lets Phase 2 converge *within* a budget PGD cannot use as well. The 40-iteration control extension argues against this, but is review-produced.
4. **Same-machine identity** of the July control and the August arms is consistent with the recorded hardware but is not stamped in run metadata.
5.  $N=1000$  is untouched. Nothing here establishes that B’s behaviour extrapolates.

## 12 Conclusions

1. B replaces Phase 1 at  $211\text{--}319\times$  less wall time in the replaced stage and reaches a *shorter* Phase 2 perimeter than the best available PGD pipeline at both  $N$ , at equal Phase 2 budget.
2. Its raw labels are connected at both  $N$  — the one validity gate B could have failed — so the readout and repair stages are unnecessary for B.
3. The advantage is attributable to the MBO dynamics, not to the balanced geodesic initialisation, by  $+2.761\%$  against a  $0.3\%$  threshold.
4. Three pre-registered predictions missed, all in B’s favour, and one was initially mis-scored in B’s *disfavour* by an instrument that did not exist.

## Provenance of the references

**Updated 2026-09-17.** Every paper cited in §1 is now held at `docs/papers/` and was read at the sections cited, not only at the abstract; the reading is recorded, with a table of cell counts, domains and constraints, in `docs/math/10-mbo-auction-dynamics/` §2. The original note stands for the two papers it covered: [13] is Shilong Hu, Hao Liu and Dong Wang (v1, 25 May 2024) — an earlier draft of this report guessed two of the three given names wrongly, from a secondary description — and [4] is Benjamin Bogosel and Edouard Oudet (v2, 7 June 2021). The consequence of the wider reading is the corrected novelty paragraph in §1.

Of the entries in `docs/math/shared/references.bib` that this report cites, every one is now either verified against a held copy or explicitly marked second-hand in its own `note` field ([22] is the one second-hand entry in the lineage). Entries this report does not cite keep the upstream note that volume and page numbers should be checked before external distribution.

## Related documents

- Plan, pre-registration and full correction history: `docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md`
- Taxonomy: `docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md`
- The instrument this report is scored through: `docs/experiments/07-phase0-shared-harness/`

- The gap B makes vacuous: docs/reference/winner\_take\_all\_partition\_gap.md

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- [3] Benjamin Bogosel and Édouard Oudet. Partitions of minimal length on manifolds. *Experimental Mathematics*, 26(4):496–508, 2017. THE foundational reference for this project: Phase 1 ( $\Gamma$ -convergence relaxation) and Phase 2 (direct perimeter minimization). Distinct from `bogosel2021longest`, which is the same authors on the OUTER problem and is cited only in the literature section of experiment 08 – do not conflate them. read `year = 2023` with no volume or pages, under the key `bogosel2023partitions`. The rest of the repository already said 2017 (`src/surfaces/double_torus.py`, `banchoff_chmutov.py`, `docs/plans/MESH_DEGENERACY_AND_NEEDLE_TRIANGLES.md`); the bibliography was the outlier. Taylor & Francis lists this as *Experimental Mathematics volume 26 (2017), issue 4*, and the article page gives “Published online: 04 Oct 2016”. Crossref carries both dates and its automatic BibTeX transform emits `year = 2016`; the DOI also encodes 2016. We cite the ISSUE year, 2017. A venue preferring online-first dating would write 2016; both refer to this same article, DOI 10.1080/10586458.2016.1223570, arXiv:1606.02873.
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- uses on the surface FEM mesh. Held; first and last page numbers read off the copy (the last page is 1613, not the 1614 some indexes give).
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  - [14] Matt Jacobs, Ekaterina Merkurjev, and Selim Esedoğlu. Auction dynamics: A volume constrained MBO scheme. *Journal of Computational Physics*, 354:288–310, 2018. AUTHORS CORRECTED 2026-08-21, verified against dblp ([journals/jcphy/JacobsME18](#)) and ScienceDirect. This entry previously read “Jacobs, Kim and Léger”, an error that originated in the archived A–E proposal ([docs/reference/phase1\\_highn\\_proposal\\_ABCDE.original.md](#)) and propagated into seven documents before being checked. Volume, pages and year were correct throughout. Note Esedoğlu is an author of both this and [esedoglu2015threshold](#).
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